Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18EC52]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

CONTROL SYSTEMS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Explain basic components of control systems? Illustrate with the help of block diagram open loop and closed control systems with suitable examples. **08**

b. List the force voltage analogy by comparing translation, rotational and electrical systems. **06**

2. a. Develop the suitable equation for steady state error for a simple closed loop system and represent the static error coefficients for the reference input, ramp input and parabolic input. **08**

b. What are the standard test signals? How are they mathematically defined? Write their Laplace transforms. **06**

3. a. Discuss the concept of stability and mention the necessary condition for stability. **05**

b. Check the stability of the system represented by the characteristics equation using RH method. **06**

$Q(s) = s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$

c. List the advantages of RH Criterion. **03**

OR

4. a. Sketch the root locus of the system having $G(s)H(s) = \frac{K}{s(s+3)(s+5)}$ Determine the system stability. **10**

b. The transfer function of a system is given by $T(s) = \frac{K(s+6)}{s(s+2)(s+5)(s^2+7s+12)}$ **04**

Determine

i. Poles

ii. Zeros

iii. Plot on s-plane

5. a. Explain the following: **08**

i. Gain Margin

ii. Phase Margin

iii. Gain crossover frequency ω_{gc}

iv. Phase crossover frequency ω_{pc}

b. Sketch the Bode Plot for $G(s) = \frac{40(s+5)}{s(s+10)(s+2)}$ **06**

OR

6. a. A Unit feedback control system has $G(s) = \frac{80}{s(s+2)(s+20)}$. Draw the Bode plot. **08**

Determine gain margin and phase margin. Comment on the stability.

b. List the steps in constructing the Bode plots. **06**

7. a. Discuss the concepts of state variable analysis? State the advantages and disadvantages of state variables. **10**

b. Construct the state model using phase variables if the system is described by the differential equation : $\frac{d^3y(t)}{dt^3} + \frac{4d^2y(t)}{dt^2} + \frac{7dy(t)}{dt} + 2y(t) = 5\ y(t)$ **04**

OR

8. a. Obtain the state model of the given electrical system shown in figure 8.a **10**

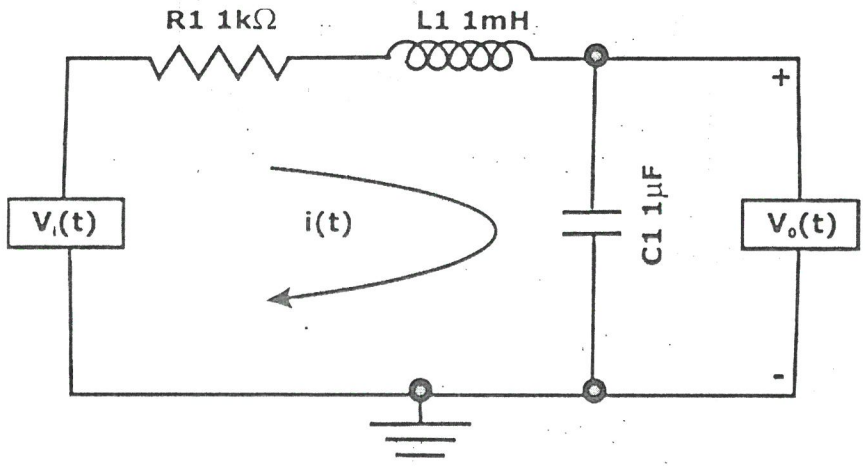

figure 8.a

b. Define the following : **04**

i) State ii) State variables iii) State vector iv) State space